Document Management System

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Child Health

Code: CH-PICU-PRT-7-Vers.1.0

Effective Date: 19/02/2024
Target Review Date: 20/02/2027

Title: Procedural Sedation & Analgesia

Analgesia and behavioural strategies should beused in all children prior to painful procedures, with sedative drugs addedwhere necessary. Sedative drugs are not a replacement for adequate analgesia,nor should they replace behavioural strategies (eg. calico dolls/play therapy)for reducing children's anxiety.

Analgesia:

Paracetamol

1. 20 mg/kg (max 90 mg/kg/day) orally or PR

Non Steroidal Anti-Inflammatory Drugs

- naproxen 5 - 10 mg/kg (max 500 mg) 12 hrlyorally or PR
- ibuprofen 2.5 - 10 mg/kg (max 600 mg) 6-8hrlyorally

Opioids

- oxycodone 0.1 - 0.2 mg/kg (max 5-10 mg)orally 4-6 hrly
- pethidine 1mg/kg i.m. or 0.25 - 0.5 mg/kg i.v.
- morphine 0.1 mg/kg i.m. or 0.05 - 0.1 mg/kgi.v.

Note:

- Use i.v. if older child/adolescent with easyintravenous access .
- Titrate boluses (ie. give 1/2 of dose first tosee effect, then repeat at 5 - 10 minute intervals as required up to maximumtotal dose).
- Respiratory depression risk - reduce doses ifcombined with sedatives. May get delayed respiratory depression after t Top use of pain.

Sucrose

Oral sucrose has been shown to reduce pain ininfants less than three months of age during minor procedures.

Oral sucrose may be given to infants duringprocedures such as blood collection, IV insertion, eye examination and lumbarpuncture.

Sucrose may be more effective if given with adummy as the dummy promotes non-nutritive sucking which contributes to calming.

Other strategies which assist in calminginfants and can be used as an adjunct to sucrose include feeding (if allowed),cuddling, and wrapping.

Other appropriate local or systemic analgesicagents should be administered as required.

Dose:

- Maximum 2mL (0.5mL for infants below 1500grams) administered orally for each procedure.
- Two minutes prior to a painful procedure,administer a small amount (around 0.25ml) of sucrose onto the infant's tongue.Offer a dummy if this is part of the infants care.
- Continue giving remainder of sucrose slowlyduring the procedure for a total dose of 2ml, until the procedure is completed.

Note:

- Sucrose is only effective if given orally.There is no effect if given via an oral or nasogastric tube.
- The addition of non-nutritive sucking enhancesthe analgesic effect of sucrose.

Specific uses of Analgesia

IV insertion

Topicalanaesthetic cream for 30 - 45 mins; distraction eg.calico dolls.

Oralsucrose combined with non-nutritive sucking for infants under three months

Lumbarpuncture

Topicalanaesthetic cream for 30 - 45 mins, 1% lignocaine with 25G needle.

Oralsucrose combined with non-nutritive sucking for infants under three months

Removalof nasal/pharyngeal foreign body

Topical phenylephrine (0.5%) & lignocaine spray (Cophenylcaine) or nebulisedlignocaine (1.0%) 1ml in 3ml 0.9% NaCl.

Ear ache

Topicalauralginotic (lie ear up for 10 mins) or 1% Lignocaine 2 drops.

Eye

Topical amethocaine 0.5% to examine; patch +/- atropine (to relieve iris spasm)

Oralsucrose combined with non-nutritive sucking for infants under three months.

Regionalanaesthesia (eg. Bier's block) - see Intravenous RegionalAnaesthesia guideline.

Systemicanalgesia - see nitrous oxide section below

Sedation :

Properties:

- anxiolytic and anterograde amnestic.
- onset is usually within minutes

Indications

- procedures such as suturing, removal of foreignbody from nose etc.

Dose

- oral / buccal 0.5 - 1.0 mg/kg.(max 15mg)
- intranasal 0.6mg/kg (max 10mg) - may have morerapid onset.
- Draw up solution (5 mg/ml) in a 1 or 2 mlsyringe. May be mixed with small volume of cordial to disguise acid taste.

Cautions

- Observe the child in Emergency until fullyalert (usually recover by 60 minutes)
- Midazolam will potentiate the effects of othersedative drugs eg. opiates.

Midazolam should not be given to children withpre-existing respiratory insufficiency, or neuromuscular problems such asmyasthenia gravis

Ketamine:

Characteristics of Ketamine Dissociative State

2. Dissociation - the patient passes into a trancelike state with the eyes open but not responding
3. Catalepsy - normal or slightly increased muscletone is maintained.
4. Analgesia - excellent analgesia is typical
5. Amnesia is usually total
6. Airway reflexes are maintained.
7. Cardiovascular state - Blood pressure and heartrate tend to increase slightly.
8. Nystagmus is typical

Potential Side Effects:

- *Unpleasant emergence phenomena - more commonbeyond mid adolescence.*
- Hypersalivation.
- Transient laryngospasm
- Transient apnea or respiratory depression
- Emesis
- Recovery agitation
- Random purposeless movements, muscle twitchingand rash are common.

Patient Selection

- Children aged over 12 months - there is anincreased risk of airway complications in children less than 12 months andparticularly less than 3 months.

- **Short painful procedures**especially those requiring immobilisation. Examples of these include:lacerations - especially of the face, and fracture reduction.

Contraindications:

- Children under 12 months (relative)
- Food within four hours.
- Chest infection /URTI or lung disease.
- History of previous airway surgery orcongenital anomaly.
- Procedures that will stimulate the posteriorpharynx.
- Cardiovascular disease including hypertension.
- Head injury with LOC, altered conscious stateor vomiting.
- Poorly controlled seizure disorder.
- Glaucoma or acute globe injury.
- Psychosis, Porphyria.
- Thyroid disease.

Procedure

Staff required:

- Airway doctor - must be approved by theEmergency Department
- Nurse
- Doctor for the procedure.

Resuscitation equipment must be readilyavailable.

Presedation

- The procedure should be explained to thecaregivers and child including an explanation of the effects of Ketamine.
- Written informed consent must be obtained.
- Baseline observations should include BP, PR, RRand O2 saturation.
- Encourage the child and parents to talk (dream)about happy topics. This helps minimise unpleasant emergence phenomena.

Adjunctive agents

- Atropine 0.02mg/kg to a maximum of 0.6mg can beused to diminish hypersalivation.
- Midazolam 0.02 mg/kg may be added to ameliorateemergence phenomena in children over 5 years old.

These may all be mixed in the same syringe forintramuscular or intravenous injection.

Administration:

IV: (with local anaesthetic cream)

- Especially useful for procedures longer than15-20 minutes
- The Ketamine dose of 1-1.5 mg/kg is givenslowly over (1-2 min) as more rapid administration is associated withrespiratory depression.
- Further incremental doses of 0.5mg/kg may begiven if sedation is inadequate or longer sedation is necessary.
- Atropine and Midazolam may be given prior to orwith the Ketamine.

IM:

- Ketamine can be safely used without i.v.access.
- 3-4 mg/kg Ketamine with atropine and Midazolammixed in the same syringe.
- A repeat dose of 2-4 mg/kg may be given after10 minutes if sedation is inadequate.

IV	IM	Route of Administration
Ease of repeat dosing, slightly faster recovery	No IV necessary	Advantages
1 minute	5 minutes	Clinical onset
10-20 minutes	15-30 minutes	Effective sedation
90-120 minutes	100 -140 minutes	Time to discharge (average)

Monitoring

- *Each patient should have pulse oximetry andcardiac monitoring, and a nurse in attendance until recovery is wellestablished.*
- Close observation of the airway and chestmovements is necessary.

Post Procedure

Recovery

- Nil orally until fully alert
- Nurse in a quiet area with minimal noise andphysical contact, allow dim lighting if possible, and do not stimulateprematurely.
- When patient is able to ambulate and verbaliseat a level consistent with their pretreatment functioning then they may bedischarged home.

Discharge Instructions

Careful family observation and no independentambulating for at least two hours.

Checked By: Dr.Said Al Hashmi
Authorized by: KHOULA JULENDA AL SAID

Back